matching by geography and industry, but the general idea is that real estate is a bundle of risk factors.

The matching procedure in Figure 14.5 is not just academic. CPPIB funds the investment in private equity by transfers from its factor portfolios of equities and bonds. This benchmark portfolio is called the Reference Portfolio, and it can be managed by twelve to fifteen people using cheap, passive index funds. The factor Reference Portfolio at writing is 65% equities and 35% bonds, with further breakdowns for domestic and foreign equities, and domestic and (hedged) foreign sovereign bonds.[21] The factors in the Reference Portfolio are chosen so that the

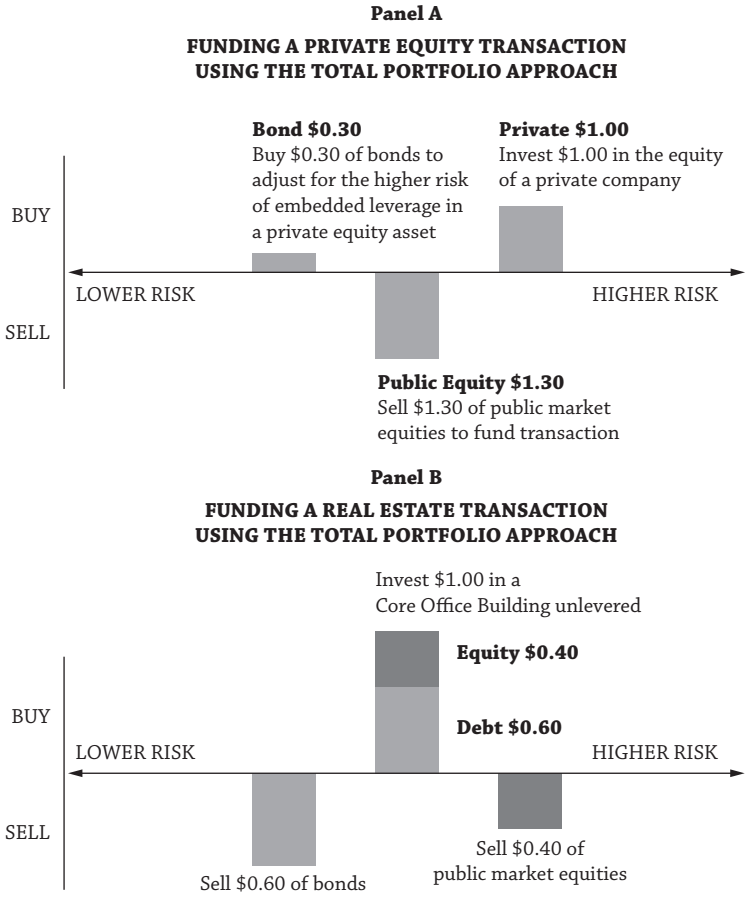

Figure 14.5

[21] The Reference Portfolio was revised on April 1, 2012. Prior to that date it was still 65% equities and 35% fixed income, but the equity portfolio was divided into domestic, foreign developed, and emerging market equities, and the bond portfolio comprised domestic fixed income, domestic real bonds, and (hedged) foreign sovereign bonds.